

TWO APPROVED NEXT STEPS

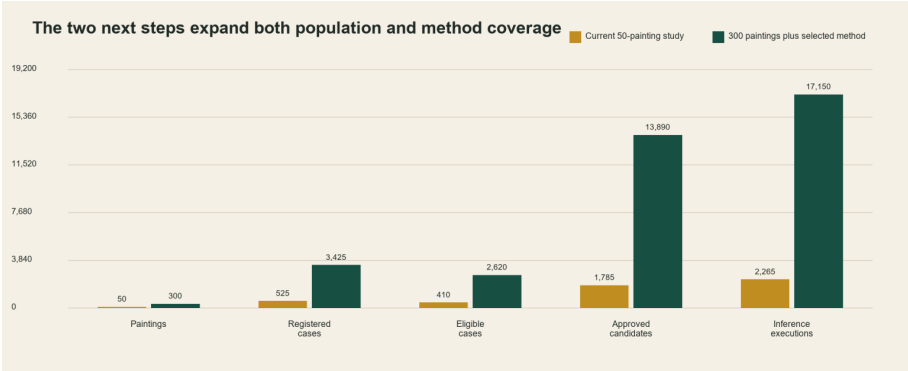
Next Steps

Execution and Resource Plan

Direct expansion to 300 paintings and one additional full-benchmark inpainting method

The next run scales the frozen design directly to 300 paintings and selects either HINT or MAT as a fourth fully evaluated method through a 12-case pilot. There is no intermediate 300-painting run.

Prepared 2026-09-06 from the validated repository evidence



300

paintings

3,425

registered cases

2,620

eligible cases

17,150

execution records

~690 h

upper serial budget

Contents and decision path

1	Locked next steps	Direct 300-painting scope plus one full-benchmark method
2	Population arithmetic	Cases, candidates, seeds, and approved evidence
3	Compute projection	Inference, metrics, maps, analysis, and reporting
4	Storage projection	Files, images, tables, checkpoints, and reserve
5	Notebook conversion	Required changes across N01-N36 plus N12A
6	Execution controls	One full run with restartable notebook gates
7	Supervisor decisions	What is locked and what can still be adjusted

Recommended reading: use pages 3-9 to approve the two scientific scope changes, pages 10-15 to approve compute and storage, and pages 16-21 as the execution checklist.

The two locked next steps

The next implementation is one direct 300-painting run with a fourth full-benchmark method selected between HINT and MAT. The current output ownership, metric separation and validation logic remain. The expanded configuration and selected-method contract are frozen before N01 is run.

Decision	Locked value	Implementation meaning	Result
Dataset	300 paintings	60 paintings in each of five broad visual categories	Direct full run
Canonical experiment	All 300 paintings	Four damage families plus one zero-damage control	1,500 cases
Focused experiments	35 paintings each	Seven paintings per visual category	Larger than a strict 6x expansion
Full restoration methods	Telea, LaMa, Stable Diffusion, plus HINT or MAT	Every restoration-eligible case after method selection	Full four-method comparison
Additional capability	HINT or MAT mask-aware transformer	Chosen by a fixed 12-case pilot before the full run	One fixed candidate per eligible case
SDXL	35 balanced cases	Seven per category; one seed; eight-hour global guardrail	Not a full benchmark method
Notebook architecture	N01-N36 plus N12A	Add the pilot-selected method before shared metric consumers	One selection pilot, not an intermediate benchmark

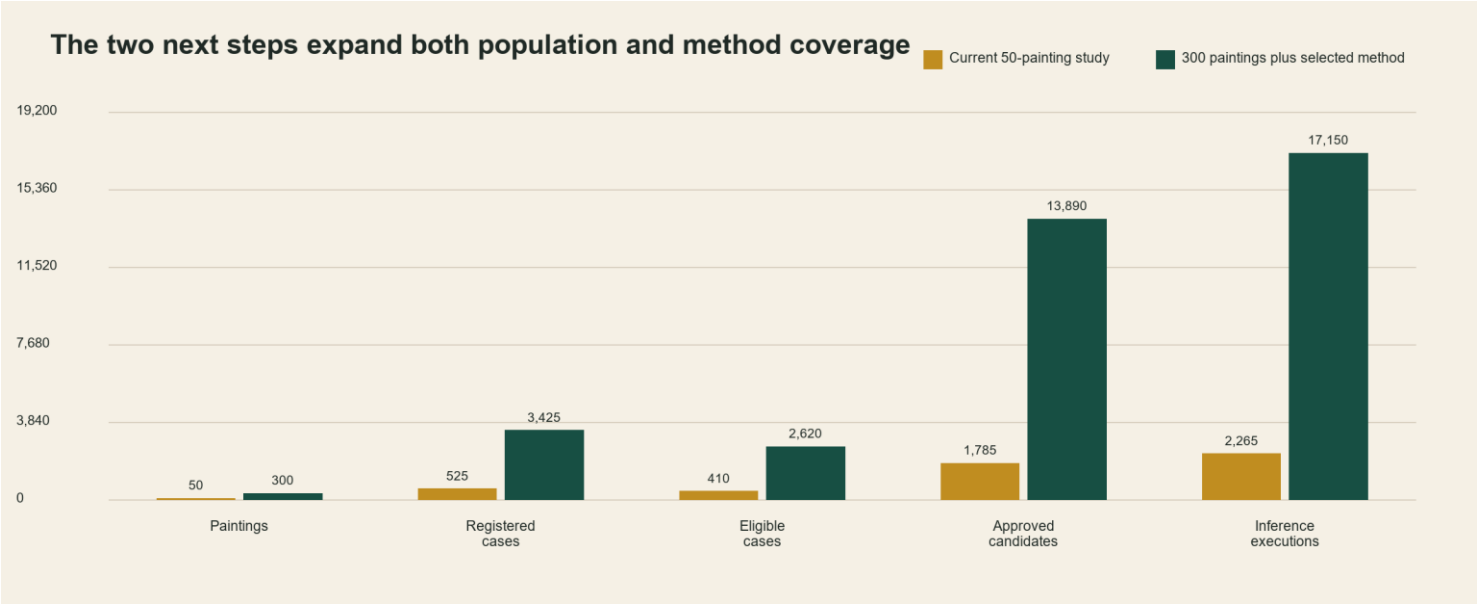
- The five grouping categories remain operational visual categories, not independently validated art-historical styles.
- All counts below are planning projections. They must remain labelled as projections until the 300-painting run passes validation.
- The 35-painting focused cohorts should be selected before generation and frozen through stable painting IDs.
- HINT and MAT remain provisional candidates until their checkpoints, licences, isolated environments and the 12-case method-selection pilot pass preflight.

Painting selection and preprocessing contract

Broad visual category	Full collection	Each focused cohort
Portrait and figure	60	7
Landscape and natural	60	7
Architecture and structured	60	7
Abstraction and surrealism	60	7
High texture and brushwork	60	7
Total	300	35

- Gather 300 rights-compatible paintings and assign stable IDs before N01.
- Record provenance, rights category, title, artist and source. Style or period, date and medium remain best-effort metadata rather than completeness gates.
- Preprocess to 768 x 768 RGB with aspect-ratio preservation, padding and recorded content bounds.
- Freeze the seven-per-category focused cohorts for damage-size, mask-robustness and synthetic-degradation experiments.
- Keep 768 x 768 as the canonical evaluation frame. Test HINT and MAT first at the largest checkpoint-compatible resolution supported by the workstation. If the selected model needs a smaller working image, apply one deterministic aspect-ratio-preserving adapter, map its result back to 768 x 768 and composite only inside the canonical mask. Freeze the adapter and report the resolution limitation before the full run.
- Do not substitute paintings after model or metric results are visible unless a documented eligibility failure requires it.

Current and projected population arithmetic



Counts grow by six for the full collection and by seven for the modestly expanded focused experiments

Population	Current	Projected	Growth	Definition
Paintings	50	300	6.0x	60 per category
Registered cases	525	3,425	6.5x	Includes non-restoration degradation cases
Restoration-eligible cases	410	2,620	6.4x	300 zero controls plus 2,320 nonzero cases
Approved candidate records	1,785	13,890	7.8x	Three existing full methods, one selected full method and bounded SDXL
Restoration execution records	2,265	17,150	7.6x	Includes selected method, prompt, uncertainty and SDXL generation

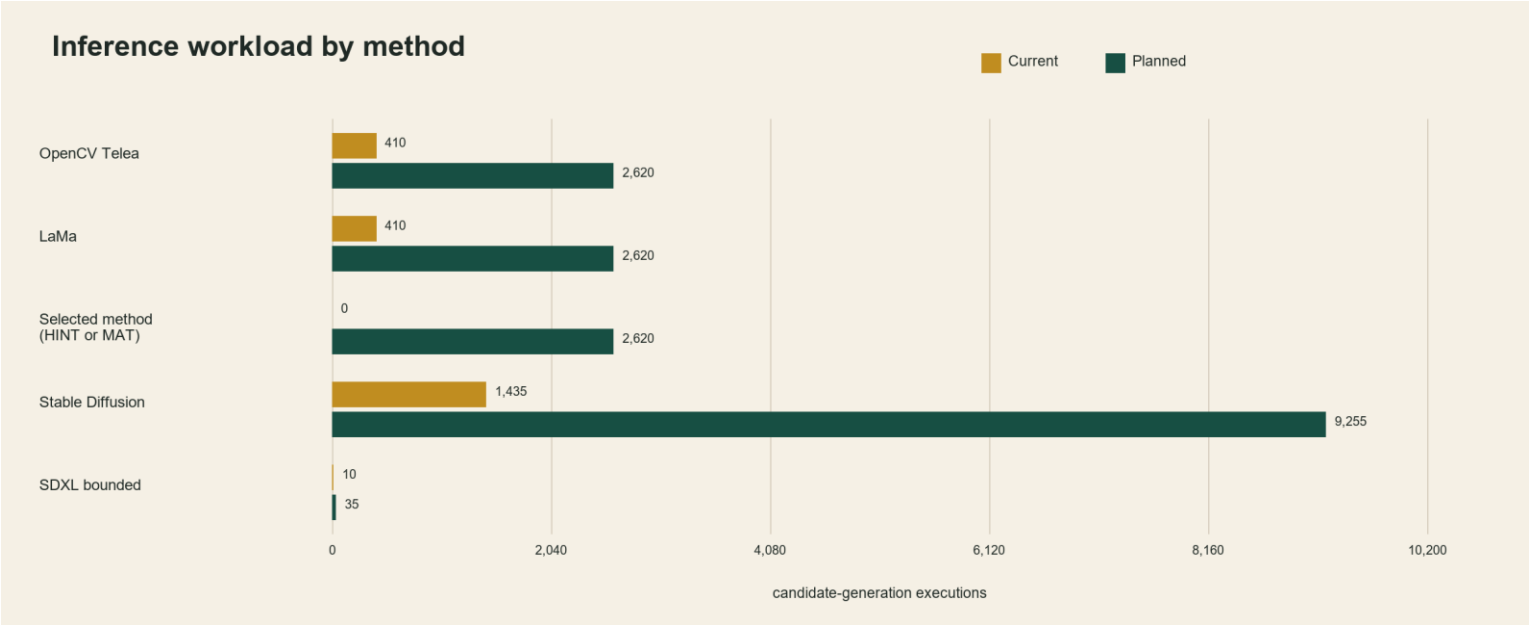
Experiment families and exact formulas

Experiment	Current arithmetic	Projected arithmetic	Painting scope	Scientific role
Canonical damage	50 x 5 = 250	300 x 5 = 1,500	All paintings	Four damage families plus zero control
Damage-size sensitivity	5 x 7 = 35	35 x 7 = 245	35 paintings	Seven nested target damage levels
Mask robustness	5 x 15 = 75	35 x 15 = 525	35 paintings	Matched mask placements or geometry conditions
Synthetic degradation registry	5 x 33 = 165	35 x 33 = 1,155	35 paintings	Single and combined degradation conditions
Eligible degradation subset	5 x 10 = 50	35 x 10 = 350	35 paintings	Mask-like removable families only

13,890 approved candidates	2,620 restoration eligible	2,320 nonzero eligible	300 zero controls
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The registered total is not the restoration total. Broad blur, global colour change and related full-image effects remain degradation evidence and do not become inpainting tasks.

Restoration workload by method



Execution workload and retained catalogue are separated because Stable Diffusion produces controlled supporting candidates

Method	Current executions	Projected executions	Projected approved records	Scope
OpenCV Telea	410	2,620	2,620	Full applicable coverage
LaMa	410	2,620	2,620	Full applicable coverage
Additional method (HINT or MAT)	0	2,620	2,620	Planned full coverage
Stable Diffusion	1,435	9,255	5,995	Full applicable coverage
SDXL bounded	10	35	35	Balanced bounded coverage

The selected HINT or MAT model adds 2,620 approved records: 2,320 nonzero inferences plus 300 zero-control identity records. It uses one frozen output per case, so the comparison gains one full method without adding a second stochastic uncertainty design. SDXL contributes 35 balanced single-seed cases: seven scenarios in each visual category.

Selecting HINT or MAT for the full benchmark

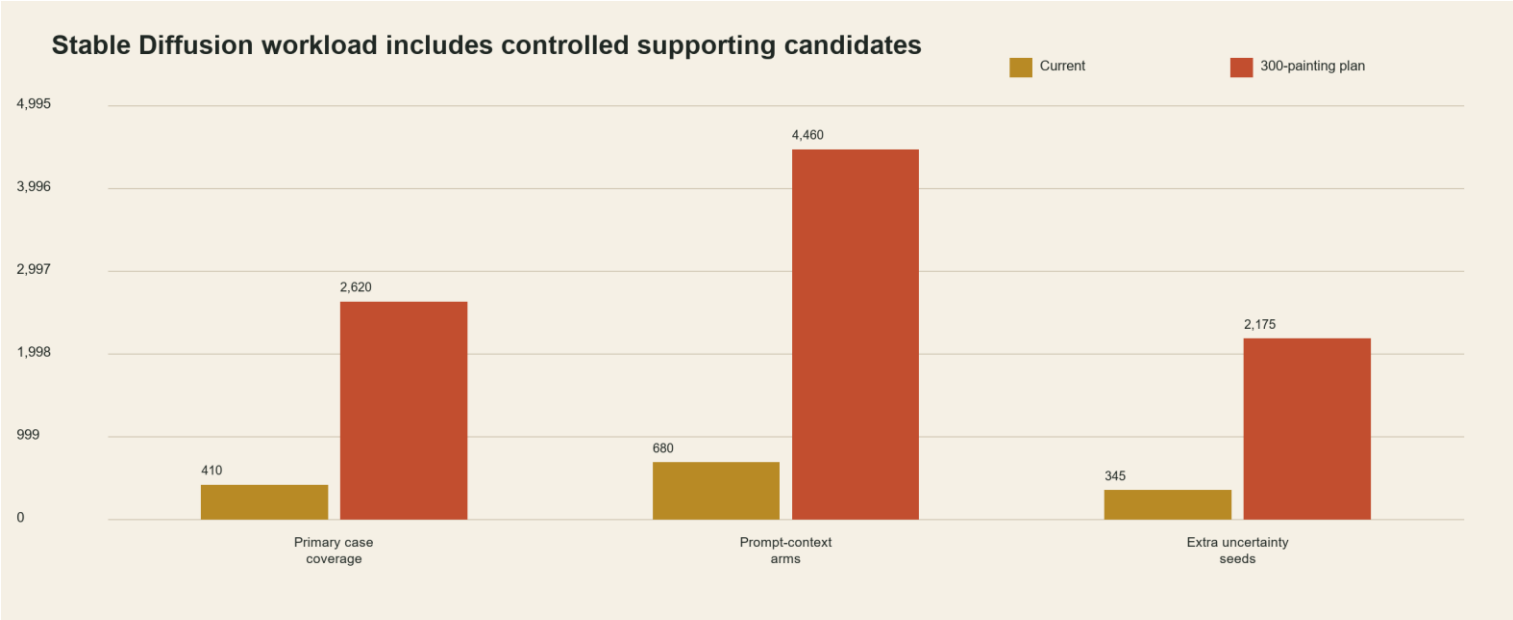
Decision factor	Candidate evidence	Planning consequence
Missing capability	HINT and MAT are mask-aware transformer methods	Adds long-range completion beyond Telea, Fourier-convolution LaMa and diffusion
Released evidence	HINT: Places2 and Dunhuang; MAT: Places 512	Use official weights without project-specific retraining
Resolution	Canonical evidence remains 768 x 768	Prefer direct 768 handling; otherwise freeze and disclose one deterministic adapter
Compatibility gates	All 12 pilot cases must complete on the workstation	Reject a model that fails memory, dimensions, mask polarity, compositing or restart checks
Selection rule	Prefer HINT after equal hard-gate success; use MAT as fallback	Do not select the model by whichever pilot metrics look better
Licence	HINT uses MIT; MAT code and models are research-use only	Record the selected checkpoint and terms in its model card

The 12-case pilot compares HINT and MAT across all five visual categories and small-to-large masks. Both candidates must load their official checkpoints, complete every case, fit GPU memory, preserve output dimensions, use the correct mask polarity and support exact outside-mask compositing. The pilot also records median and high-percentile runtime, deterministic behaviour, restart safety and any resolution adapter.

The additional full-method branch raises the projected approved catalogue from 11,270 to 13,890 records, a 23.2% increase. Prefer HINT if it passes the hard gates and handles the canonical frame with less lossy adaptation; otherwise use MAT. Pilot metric results are documented but do not select the winner. One declared candidate identity is frozen for the main comparison, and the method does not add uncertainty groups.

Primary sources: [HINT paper](#) | [HINT implementation](#) | [MAT paper](#) | [MAT implementation](#)

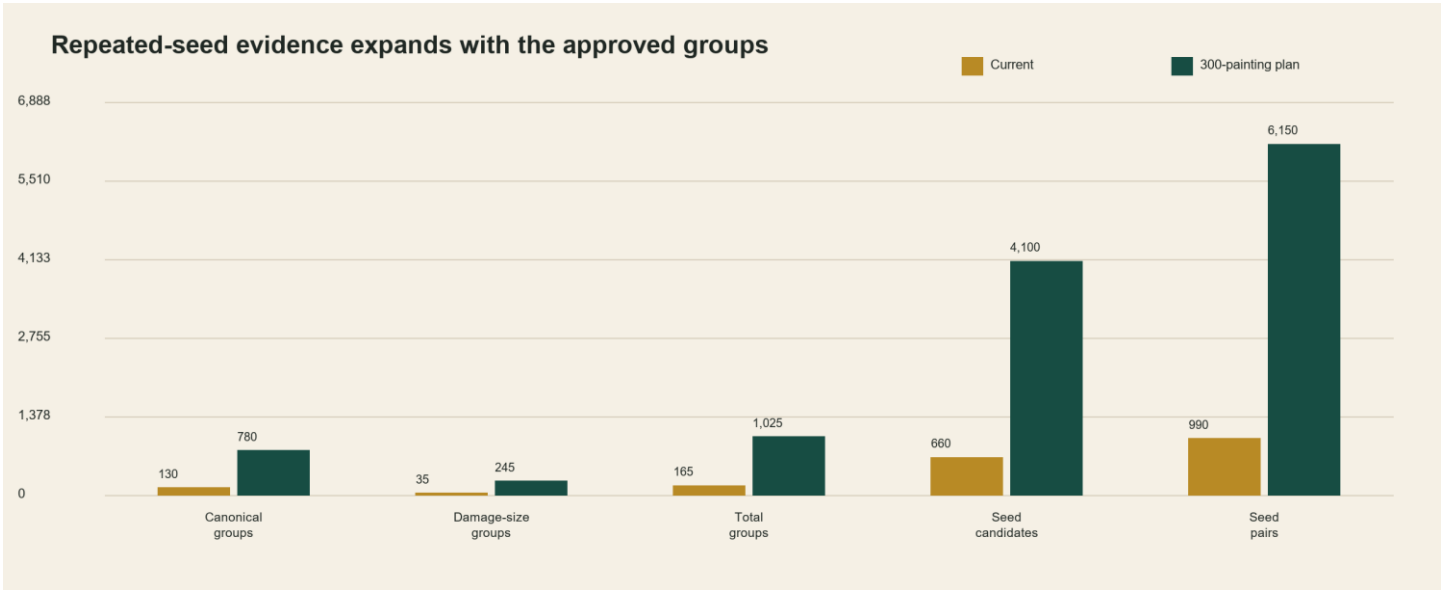
Stable Diffusion workload anatomy



Projected Stable Diffusion work retains primary, prompt-context and repeated-seed roles

Generation role	Current	Projected	Meaning
Primary case coverage	410	2,620	One declared primary candidate for every eligible case
Prompt-context arms	680	4,460	Category, artist, combined context and scratch-aware ablations
Additional uncertainty seeds	345	2,175	Canonical and damage-size extensions beyond primary seeds
Total Stable Diffusion executions	1,435	9,255	Full generation workload
Approved Stable Diffusion catalogue	955	5,995	Evidence retained under the frozen selection rules

Repeated-seed uncertainty design

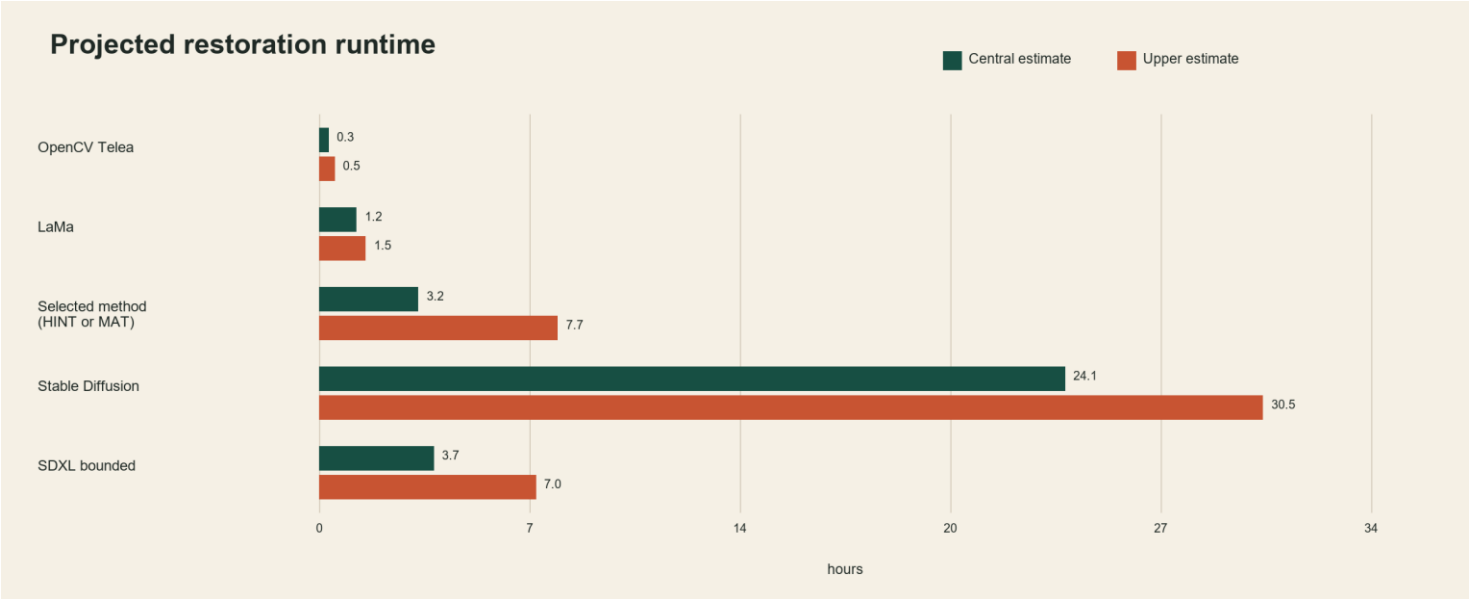


Four seeds are retained within every approved uncertainty group

Evidence unit	Current	Projected	Rule
Canonical prompt-specific groups	130	780	6x full-collection scaling
Damage-size groups	35	245	7x focused-cohort scaling
Complete uncertainty groups	165	1,025	Fixed case, prompt arm and generation configuration
Four-seed candidates	660	4,100	Seeds 2026, 2027, 2028 and 2029
Unique unordered seed pairs	990	6,150	Six pairs per complete four-seed group

Variation remains empirical candidate disagreement. It can prioritize inspection, but it is not calibrated confidence and agreement does not establish correctness.

Model inference runtime

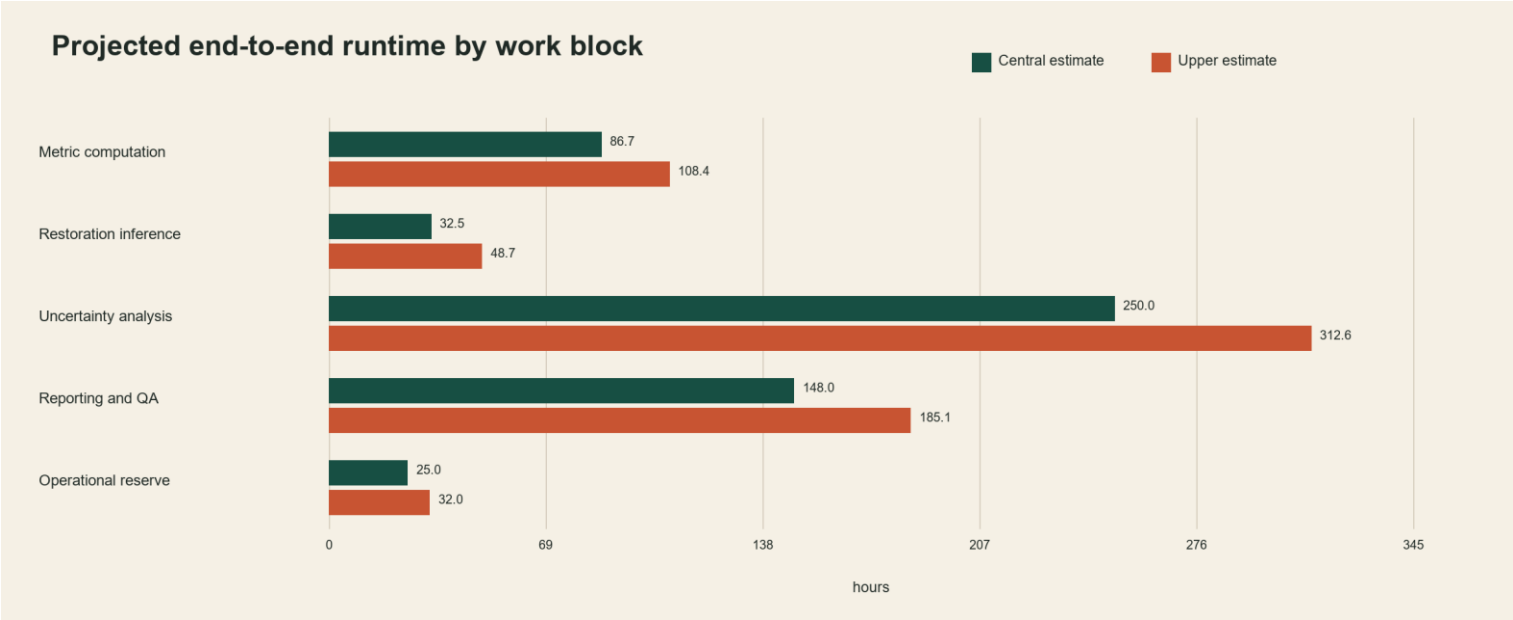


The upper budget uses observed high-percentile rates plus operational headroom

Method	Execution records	Per generated case	Central total	Upper budget	Basis
OpenCV Telea	2,620	0.45 s	20 min	32 min	Recorded workstation
LaMa	2,620	1.60 s	1.2 h	1.5 h	Recorded workstation
Additional method (HINT or MAT)	2,620	5.0 s provisional	3.2 h	7.7 h	Pilot allowance
Stable Diffusion	9,255	9.37 s	24.1 h	30.5 h	Recorded workstation
SDXL bounded	35	378.5 s	3.7 h	7.0 h	Balanced bounded run

Total restoration inference is about 32.5 serial hours at observed and provisional rates. Reserve 48.7 hours. The selected HINT or MAT model retains a 3.2-hour central and 7.7-hour upper allowance until the pilot replaces it with measured rates. The 35-case SDXL branch requires about 3.7 hours at the observed rate and receives a seven-hour upper allowance.

End to end workstation runtime



Saved diagnostics and downstream analysis are the largest projected compute blocks

Notebook responsibility block	Central estimate	Upper budget	Upper 24-hour equivalent
Foundation and datasets	86.7 h	108.4 h	4.5 days
Restoration inference	32.5 h	48.7 h	2.0 days
Metrics maps and uncertainty	250.0 h	312.6 h	13.0 days
Comparison sensitivity and XAI	148.0 h	185.1 h	7.7 days
Reports dashboard and package	25.0 h	32.0 h	1.3 days
Total	542.2 h	686.8 h	28.6 continuous days

The upper schedule is deliberately conservative. Non-model values use manifest start-to-finish spans, so they include loading, validation, serialization and some orchestration overhead. They are planning ceilings, not confidence intervals.

Metric and map runtime hotspots

Notebook	Observed span	Workload factor	Projected span	Dominant work
N13	3.62 h	7.78x	28.2 h	Classical metric-region rows
N14	0.48 h	7.78x	3.8 h	LPIPS contiguous crops
N15	0.55 h	7.78x	4.3 h	CLIP and DINOv2 embeddings and similarities
N16	12.11 h	7.78x	94.2 h	Difference maps and spatial diagnostics
N17	7.05 h	7.78x	54.8 h	Colour, texture, direction and seam maps
N18	0.11 h	6.00x	0.6 h	Canonical scalar and pairwise uncertainty
N19	3.43 h	6.00x	20.6 h	Canonical uncertainty maps and overlays
N20	3.64 h	7.78x	28.3 h	Semantic and structural maps
N22	1.58 h	7.00x	8.8 h	Damage-size seed extension and uncertainty evidence excluding inference counted earlier

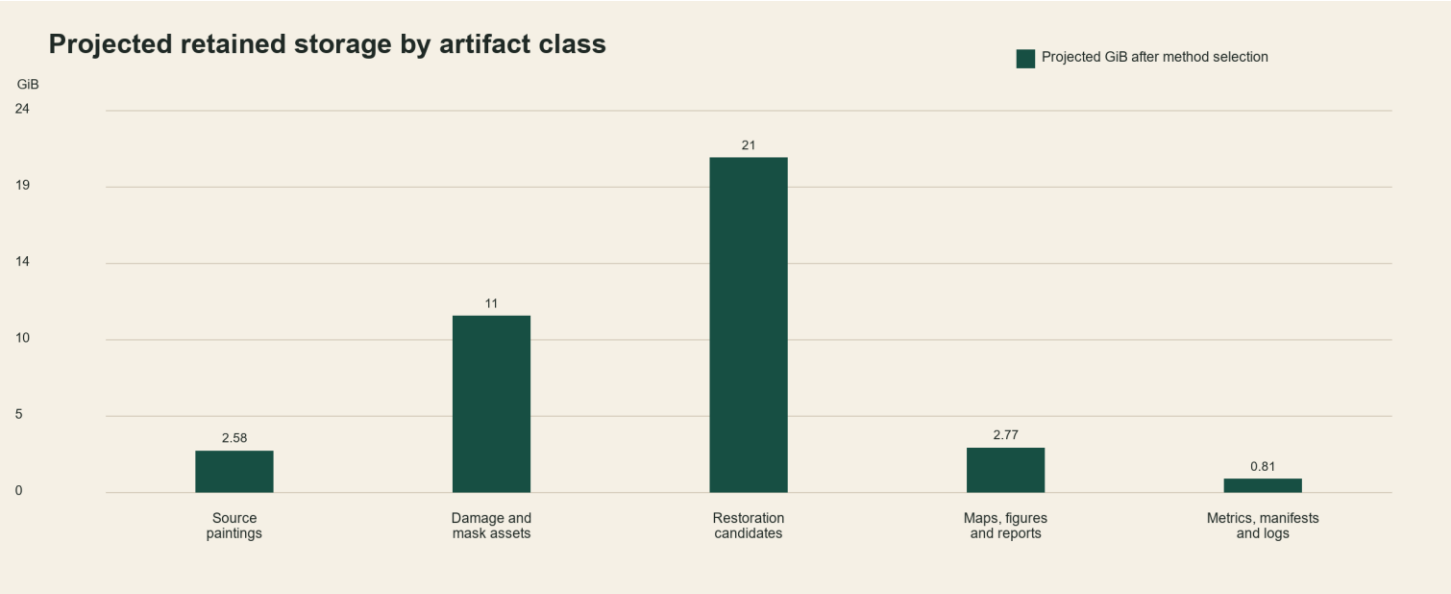
N16 and N17 alone account for roughly 149 projected hours because they materialize many image-space and local-consistency diagnostics for the larger four-method catalogue. They must be checkpointed as carefully as model inference.

Calendar interpretation

View	Calculation	Planning result	Interpretation
Ideal continuous machine time	686.8 h / 24	About 28.6 days	No downtime, no manual review
Twelve active machine hours per day	686.8 h / 12	About 57.2 active days	More realistic for supervised local execution
Recommended calendar allowance	Direct run plus review and reruns	Eight to ten weeks	Still one full design, not phased experimentation
GPU inference block	Serial and checkpointed	About 2.0 days upper	Includes the pilot-selected method and 35-case SDXL branch
Metric and analysis block	Serial upper budget	About 20.7 days continuous	Dominant schedule risk

- Do not promise a completion date from the 32.5-hour model estimate alone.
- Before the full run, compare HINT and MAT on a fixed 12-case method-selection pilot spanning all five visual categories and small-to-large masks. The pilot qualifies one method for full coverage and remains separate from the main 300-painting evidence.
- The schedule assumes the current workstation and software stack. A changed GPU, CUDA build or storage device requires a new observed-rate record.
- Parallel CPU work may shorten elapsed time, but the plan does not depend on unvalidated concurrency gains.

Projected output storage and file volume



Central projected storage by notebook responsibility block

Block	Projected size	Files	Image files	Tabular rows
Foundation and datasets	2.58 GiB	7,504	7,214	32,591
Restoration generation	11.06 GiB	16,431	16,288	20,982
Metrics maps and uncertainty	21.00 GiB	115,641	115,327	3,744,914
Comparison sensitivity and XAI	2.77 GiB	1,973	1,526	1,614,880
Reports dashboard and package	0.81 GiB	1,256	293	371,917
Central projected total	38.3 GiB	142,805	140,938	5,760,196

Storage reserve and output policy

Storage class	Projection basis	Recommended reserve	Rule
Final notebook-owned outputs	38.3 GiB central	Reserve 55 GiB	Includes selected-method maps, reports, tables and manifests
Transient checkpoints and rerun overlap	Not retained	Reserve 30 GiB	Atomic checkpoints, temporary images and duplicate exports
Model caches and environments	Excluded from outputs	Reserve 50 GiB	Includes isolated HINT and MAT pilot stacks and checkpoint caches
Operating margin	Free space	Reserve 45 GiB	Prevents late-run failure during map or report export
Recommended free capacity before N01	Combined planning envelope	180 GiB	Upper-side operational target

- Keep outputs/inventory as the sole global output exception; every other artifact remains inside its producer notebook's output root.
- Persist canonical long-form tables and rebuild disposable summaries instead of multiplying CSV files.
- Retain complete candidate and map indexes even when reports display representative subsets.
- Clear a notebook's output directory before a deliberate full rerun, but never delete upstream evidence during a downstream rerun.
- Record hashes, dimensions, row counts and producer notebook IDs in the refreshed inventory.

Notebook conversion map N01 to N12

Notebooks	Required scale change	Completion gate
N01-N02	Replace 50-painting config with balanced 300-painting manifest; verify rights, metadata, 768 x 768 outputs and content bounds	300 verified and preprocessed paintings
N03-N04	Generate five canonical cases per painting with deterministic mask seeds and damage audit	1,500 canonical cases
N05	Freeze 35-painting cohort and generate seven nested damage levels	245 damage-size cases
N06	Freeze 35-painting cohort and generate 15 matched mask conditions	525 robustness cases
N07	Generate full 33-condition degradation registry for 35 paintings; retain ten eligible removal conditions per painting	1,155 registered; 350 eligible
N08	Rebuild normalized case registry, eligibility, region policy and four-method applicability	3,425 registered; 2,620 eligible
N09-N10	Run Telea and LaMa once for every eligible case with progress every ten and checkpointed manifests	2,620 outputs per method
N11	Run 2,620 primary SD cases plus scaled prompt-context and canonical uncertainty roles	8,520 N11 executions
N12	Run the balanced 35-case SDXL design with the existing per-case timeout and an eight-hour global guardrail	35 feasibility candidates; seven per visual category
N12A	Run one fixed candidate from the pilot-selected HINT or MAT model for every eligible case; map and composite outputs into the canonical 768 x 768 frame	2,620 approved records; 2,320 generated outputs

Notebook conversion map N13 to N22

Notebook	Required scale change	Completion gate
N13	Chunk classical metric-region computation over the full four-method approved candidate catalogue	Finite rows, approved regions and expected denominators
N14	Checkpoint LPIPS by candidate and contiguous crop	No sparse-pixel LPIPS inputs
N15	Cache CLIP and DINOv2 embeddings once per unique image/crop	Embedding uniqueness and path audit
N16	Batch difference maps and spatial diagnostics; avoid regenerating unchanged base images	Complete map index and no temporary leakage
N17	Batch colour, texture, directional and seam diagnostics with restartable map writes	Local evidence coverage for every applicable candidate
N18	Expand canonical prompt-specific uncertainty groups from 130 to 780	Exactly four seeds per group and six pairs
N19	Generate indexed numeric maps and visual overlays for all canonical uncertainty groups	No combined uncertainty score
N20	Cache semantic and structural features and generate only declared maps	Applicable local-feature coverage
N21	Rebuild paired model comparisons using painting-level independence	Four-method comparison; SDXL separate
N22	Generate the remaining 735 damage-size seed extensions and rebuild 245-group uncertainty evidence	Complete four-seed size groups

Notebook conversion map N23 to N36

Notebooks	Required scale change	Completion gate
N23-N25	Recompute damage-size, mask and eligible degradation trajectories for the 35-painting cohorts	Grouped estimates retain painting as the independent unit
N26	Rebuild paired/grouped tests, effect sizes, intervals and multiplicity correction	No candidate-row pseudo-replication
N27	Recompute transparent failure assignments and conservative review guidance	Rules remain inspectable and non-expert
N28	Repeat metric and region policy ablations; checkpoint the longest ranking-stability blocks	No universal combined score
N29	Rebuild explanations and retrieval over every approved candidate; enlarge retrieval indexes without treating neighbours as truth	Complete catalogue plus selected visual units
N30	Replace projections with observed 300-painting compute, storage and five-model-card evidence	Executed values labelled observed
N31	Regenerate four full-method reports plus the separate SDXL feasibility report	Self-contained embedded assets
N32	Generate 300 painting reports and 300 selected case reports	Complete indexes, representative report visuals
N33	Regenerate the thesis-level report and exact evidence catalogue	Claims linked to denominators and limitations
N34-N35	Rebuild dashboard package, validate filters, exact tables, local assets and deployment	No inference in dashboard; zero blocking checks
N36	Regenerate supervisor and reproducibility package with checksums	Portable validated package

Single full run execution sequence

This sequence is not a pilot ladder. All 300 paintings and all frozen experiment cohorts are active from the first configuration. The boundaries below are restart and verification gates inside one full run.

Order	Notebook range	Action	Gate
1	N01-N08	Freeze data, masks, cases, eligibility and regions	Do not start restoration until every expected count and path passes
2	N09-N12A	Generate four full-method outputs and balanced 35-case SDXL evidence	Checkpoint every ten outputs; preserve completed candidates
3	N13-N17	Compute candidate-level metrics and local maps	Chunk long jobs and validate coverage before uncertainty
4	N18-N22	Compute repeated-seed scalar and spatial evidence	Require four seeds and six unique pairs per group
5	N23-N29	Run sensitivity, statistics, flags, ablations and XAI	Use painting-level independence and complete candidate indexes
6	N30-N36	Replace projections, generate reports/dashboard/package	Finish only after artifact, manifest and inventory checks pass

Checkpoint and restart rules

Control	Required behaviour
Progress	Print progress every ten cases or candidates for generation and other long loops.
Checkpoint	Persist normalized checkpoints at declared intervals and verify them before resuming.
Atomic writes	Write to a temporary sibling, close handles, then replace; include a Windows-safe retry path.
Resume identity	Resume only when case ID, model, prompt, seed, configuration fingerprint and output hash agree.
Timeouts	Keep the 25-minute SDXL per-case timeout and eight-hour global budget; record unexecuted cases rather than silently extending scope.
Metrics	Checkpoint N16, N17, N20 and N28 by stable work unit; do not recompute completed maps after a downstream failure.
Validation	A non-blocking warning must be explained. A blocking failure stops the notebook and prevents downstream publication.
Rerun	Delete only the current notebook's owned output root before an intentional clean rerun.

Pre run capacity and integrity checklist

- At least 180 GiB free on the workspace volume before N01 starts.
- The official HINT and MAT checkpoints, licences and isolated environments are verified before the method-selection pilot.
- A fixed 12-case HINT-versus-MAT pilot records completion, load time, median and high-percentile latency, peak VRAM, output determinism, mask handling and transform fidelity. The chosen method replaces the provisional 5-to-12-second allowance with measured rates before the full run.
- Model caches are complete and remain outside notebook-owned outputs.
- The 300-row painting manifest has 60 paintings per broad visual category and no duplicate stable IDs.
- Each 35-painting focused cohort has seven paintings per category and is frozen in configuration.
- All N01-N36 notebooks and N12A read scope and the selected additional method from configuration rather than hard-coded counts.
- Expected-count assertions are updated before execution, not relaxed after failures.
- Stable Diffusion uses seeds 2026-2029 for every declared uncertainty group.
- Report generators retain complete numeric and path indexes while embedding representative visuals.
- The inventory is refreshed after every completed notebook and at the end of N36.
- Git commits occur after validated notebook boundaries; large generated outputs follow the existing repository policy.

180 GiB

free capacity

4 seeds

uncertainty groups

35 cases

bounded SDXL

0

blocking publication checks

Supervisor decisions and adjustable controls

Status	Decision	Consequence
Locked	300 paintings in one direct run	Defines the next executed benchmark
Locked	Five balanced broad visual categories	60 paintings per category
Locked	35 paintings per focused experiment	Seven paintings per category
Locked	Telea, LaMa and Stable Diffusion plus one selected HINT or MAT model on all eligible cases	Full four-method comparison after the selection pilot
Provisional until pilot	One fixed HINT or MAT candidate per eligible case	HINT preferred after equal hard-gate success; MAT fallback
Locked	Four Stable Diffusion seeds for declared uncertainty groups	Empirical disagreement design
Locked	SDXL uses 35 balanced single-seed cases	Seven scenarios per visual category; feasibility evidence, not full ranking
Adjustable before N01	Which paintings enter the focused and SDXL selections	Must be frozen before generation
Adjustable before N01	Storage location and run calendar	Does not alter scientific scope
Adjustable before N01	Selected-method adapter and working resolution	Freeze the least lossy compatible transform before full generation
Gate before N01	HINT-versus-MAT method-selection pilot	Replace provisional runtime, environment and adapter assumptions
Not adjustable after results	Seeds, candidate selection and report examples	Prevents result-driven redesign

Final planning summary

3,425 registered cases	13,890 approved records	1,025 uncertainty groups	~38.3 GiB central outputs	~690 h upper serial budget
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- The design expands directly to 300 paintings; it does not introduce a second pilot or intermediate benchmark.
- The focused experiments grow to 35 paintings each, which is deliberately larger than a strict sixfold replication of the current five-painting cohorts.
- All four fully evaluated restoration methods cover every applicable case. The 12-case pilot selects HINT or MAT as the additional mask-aware transformer method. SDXL expands to a balanced 35-case, single-seed feasibility branch.
- The largest resource risk is not restoration inference. Metric maps, local-consistency diagnostics, semantic evidence and downstream analyses account for most projected workstation time and files.
- Reserve about 690 serial workstation hours and 180 GiB of free capacity. These are upper-side operational budgets, not executed results or statistical intervals.
- The existing N01-N36 structure remains viable with a new N12A selected-method producer if hard-coded counts are replaced with configuration-driven expectations and long metric notebooks gain production-grade checkpointing.

Recommended decision: approve the 300-painting configuration and 35-case SDXL branch, run the HINT-versus-MAT selection pilot, freeze one full method and then prepare the N01-N36 plus N12A rerun.

[Current validated dashboard](#)